

5 public releases: OU VLE, Toledo, three Moodle sites



one adapter per release

4 weekly counters per student:
clicks, active days, content clicks, social clicks



cut at week $L/3$ into 7 leakage-safe features

63 cohorts, 35,529 distinct students
one course, one period, one institution each



*gradient boosting per source cohort;
logistic regression, random forest as checks*

3,969 source–target pairs
D0 within cohort, D1 same course next year,
D2 other course, D3 other institution



*permutation, SHAP and drop-column
on the same fitted models*

agreement between feature rankings (Kendall τ)
read against noise floor, ceiling, random baseline